

CBSE Class 12 Pe — Half-Yearly Predicted Paper (Set 1 of 10)

SECTION A: Objective Type Questions (18 Marks: 18 x 1)

All questions are compulsory. Consists of MCQs and Assertion-Reason questions.

Q1. What is the formula to calculate the total number of matches in a single league tournament? (A) $N - 1$ (B) $\frac{N(N-1)}{2}$ (C) $N(N - 1)$ (D) $\frac{N+1}{2}$ *Unit 1: Management of Sporting Events | Tag: Very High***

Q2. Which one of the following is NOT a component of the Female Athlete Triad? (A) Amenorrhea (B) Eating disorder (C) Osteoporosis (D) Scoliosis *Unit 2: Children & Women in Sports | Tag: Very High***

Q3. Which of the following asanas is highly recommended to be performed immediately after eating meals to cure obesity and improve digestion? (A) Tadasana (B) Chakrasana (C) Vajrasana (D) Halasana *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | Tag: Very High***

Q4. The Deaflympics Games were first organized in the year _____. (A) 1896 (B) 1924 (C) 1960 (D) 1951 *Unit 4: Physical Education & Sports for CWSN | Tag: Very High***

Q5. Which of the following vitamins are considered 'Fat-Soluble' and are stored in the body's fatty tissues? (A) Vitamin B & C (B) Vitamin D & K (C) Vitamin B & E (D) Vitamin C & A *Unit 5: Sports & Nutrition | Tag: Very High***

Q6. Who is credited with developing the Harvard Step Test to measure cardiovascular fitness? (A) Brouha (B) Miller (C) Coubertin (D) Rikli and Jones *Unit 6: Test and Measurement in Sports | Tag: Very High***

Q7. A severe bone injury where the bone is crushed or broken into three or more pieces is medically termed as a: (A) Greenstick fracture (B) Transverse fracture (C) Comminuted fracture (D) Compound fracture *Unit 7: Physiology & Injuries in Sport | Tag: Very High***

Q8. "The more force one exerts on the downward bounce, the higher the basketball bounces into the air." Which law is this statement referring to? (A) Newton's First Law (Law of Inertia) (B) Newton's Second Law (Law of Acceleration) (C) Newton's Third Law (Law of Action & Reaction) (D) Law of Gravitation *Unit 8: Biomechanics and Sports | Tag: Very High***

Q9. Which of the following is an objective of Intramural tournaments? (A) To achieve high performance at international levels. (B) To provide an opportunity for mass participation of students within the institution. (C) To participate in inter-school competitions. (D) To promote extreme competitiveness among different schools. *Unit 1: Management of Sporting Events | Tag: Very High***

Q10. The spinal postural deformity commonly known as "hunchback", characterized by an exaggerated outward curvature of the thoracic region, is: (A) Kyphosis (B) Lordosis (C)

Scoliosis (D) Flat foot *Unit 2: Children & Women in Sports | Tag: Very High***

Q11. To be eligible to participate in the Deaflympics, an athlete must have a hearing loss of at least __ in their better ear. (A) 25 dB (B) 40 dB (C) 55 dB (D) 70 dB *Unit 4: Physical Education & Sports for CWSN | Tag: High***

Q12. Scurvy disease is primarily caused due to the severe deficiency of: (A) Vitamin A (B) Vitamin B (C) Vitamin C (D) Vitamin D *Unit 5: Sports & Nutrition | Tag: Very High***

Q13. In the SAI Khelo India Fitness Test battery for the 5-8 years age group, which specific test is used to measure static balance? (A) Plate Tapping Test (B) 50m Dash (C) Partial Curl-up (D) Flamingo Balance Test *Unit 6: Test and Measurement in Sports | Tag: Very High***

Q14. The amount of blood pumped by the left ventricle of the heart in one beat (stroke) is called: (A) Cardiac Output (B) Stroke Volume (C) Tidal Volume (D) Residual Volume *Unit 7: Physiology & Injuries in Sport | Tag: High***

Q15. The optimal angle of release to achieve the maximum horizontal distance for a projectile (like a javelin throw) is theoretically: (A) 30° (B) 45° (C) 60° (D) 90° *Unit 8: Biomechanics and Sports | Tag: Very High***

Q16. Assertion (A): The Basal Metabolic Rate (BMR) is the minimum number of calories needed to maintain body function at resting condition. **Reason (R):** A person who does not engage in any physical work still requires constant energy for the functioning of their internal organs like the heart and lungs. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 5: Sports & Nutrition | Tag: Very High***

Q17. Assertion (A): Friction is often considered a "necessary evil" in sports. **Reason (R):** While friction can slow down athletes and cause wear and tear, it is absolutely essential for generating grip, such as a sprinter wearing spiked shoes to prevent slipping. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 8: Biomechanics and Sports | Tag: Very High***

Q18. Assertion (A): The knock-out tournament is an elimination tournament. **Reason (R):** In a knock-out tournament, the winner of each match advances in the tournament, and the defeated team is immediately eliminated from the competition. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 1: Management of Sporting Events | Tag: Very High***

SECTION B: Very Short Answer Type (10 Marks: Attempt any 5 out of 6 x 2)

Word limit: 60-90 words.

Q19. What is a 'Bye' in tournament fixtures? Differentiate it briefly from 'Seeding'. *Unit 1: Management of Sporting Events | Tag: High***

- Q20.** State any two corrective measures for the postural deformity known as 'Knock Knees' (Genu Valgum). *Unit 2: Children & Women in Sports | Tag: Very High***
- Q21.** Briefly explain the health benefits of performing 'Pawanmuktasana'. *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | Tag: High***
- Q22.** What do you understand by the concept of 'Inclusion' in physical education? *Unit 4: Physical Education & Sports for CWSN | Tag: Very High***
- Q23.** Distinguish clearly between a 'Sprain' and a 'Strain'. *Unit 7: Physiology & Injuries in Sport | Tag: Very High***
- Q24.** Define the 'Centre of Gravity'. How does a lower Centre of Gravity affect an athlete's stability? *Unit 8: Biomechanics and Sports | Tag: High***
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SECTION C: Short Answer Type (15 Marks: Attempt any 5 out of 6 x 3)

Word limit: 100-150 words.

- Q25.** Differentiate clearly between Intramural and Extramural competitions by providing any three distinct points of comparison. *Unit 1: Management of Sporting Events | Tag: Very High***
- Q26.** What is 'Flat Foot'? Discuss its symptoms and suggest any two corrective physical activities. *Unit 2: Children & Women in Sports | Tag: Very High***
- Q27.** Enlist the "pitfalls of dieting" that athletes often fall into while trying to lose weight rapidly. Mention any three points. *Unit 5: Sports & Nutrition | Tag: Very High***
- Q28.** Detail the procedure and the short formula used to calculate the Fitness Index in the Harvard Step Test. *Unit 6: Test and Measurement in Sports | Tag: Very High***
- Q29.** Explain how the composition of muscle fibers (slow-twitch red fibers and fast-twitch white fibers) determines the speed and endurance of an athlete. *Unit 7: Physiology & Injuries in Sport | Tag: High***
- Q30.** Explain the three classes of levers (Class I, II, and III) giving one anatomical or sports example for each. *Unit 8: Biomechanics and Sports | Tag: Very High***
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SECTION D: Case-Study Based Questions (12 Marks: 3 x 4)

All 3 questions are compulsory. Internal choice is provided in one sub-part of each.

- Q31.** ABC School is one of the reputed schools in their location for the sports facilities it provides. Keeping that in consideration, the CBSE Sports cell has given them the responsibility of conducting the CBSE Football cluster. 35 teams have sent their entry for participation. **(i)** Due to a large number of teams willing to participate, the school is conducting the competition by which type of elimination fixture? **(ii)** What will be the total number of matches played in the tournament? **(iii)** What will be the total number of rounds in this fixture? **(iv)** Calculate the total number of byes to be given in this fixture. **OR** Write the

formula to calculate the number of byes in the lower half if the total number of teams is odd.

*Unit 1: Management of Sporting Events | Tag: Very High***

Q32. Mr. Lakshman, aged 65 years, worked as a civil engineer. He had to walk and climb a lot as part of his job. After retirement, he is experiencing difficulty in doing certain chores which involve physical movement and flexibility. His doctor advises a physical fitness evaluation. **(i)** Which specific test battery would you recommend to check Mr. Lakshman's fitness? **(ii)** Which specific test from this battery should be conducted to assess his Lower Body Flexibility? **(iii)** Which specific test evaluates his Upper Body Strength? **(iv)** The '8-foot up & go test' is performed to assess which fitness component? **OR** Mention the weight of the dumbbell used for men in the Arm Curl Test. *Unit 6: Test and Measurement in Sports | Tag: Very High***

Q33. Neha is deaf and mute by birth. She communicates through lip movements and gestures. The other students in the class hesitate to talk and play with her. Whenever there is a team activity, no one is willing to include her, making her feel isolated. Raghav approached their PE teacher, Sarita, about this issue. Sarita suggested ways Neha can be included in physical activities. **(i)** What kind of disability is Neha facing? **(ii)** The process of educating such children in the same environment as mainstream peers is called __. **(iii)** Name the specific international organization that promotes sporting events exclusively for athletes with hearing impairments. **(iv)** Suggest any one strategy the PE teacher can use to make physical activities accessible for her. **OR** Mention the minimum hearing loss limit (in dB) required to participate in Deaflympics. *Unit 4: Physical Education & Sports for CWSN | Tag: Very High***

SECTION E: Long Answer Type (15 Marks: Attempt any 3 out of 4 x 5)

Word limit: 200-300 words.

Q34. What is a League Tournament? Draw a fixture of nine (9) teams on the basis of a single league tournament using the Cyclic Method. *Unit 1: Management of Sporting Events | Tag: Very High***

Q35. Define the 'Female Athlete Triad'. Explain its three core components in detail and suggest preventive measures to avoid this syndrome. *Unit 2: Children & Women in Sports | Tag: Very High***

Q36. The dietary requirements of an athlete change drastically depending on the competition phase. Explain in detail the Pre-competition, During-competition, and Post-competition dietary strategies for an athlete to ensure peak performance and fast recovery. *Unit 5: Sports & Nutrition | Tag: High***

Q37. Define 'Friction' in sports. Distinguish clearly between Static, Sliding, and Rolling Friction. Elaborate on the specific methods used to purposefully increase and decrease friction in various sports. *Unit 8: Biomechanics and Sports | Tag: Very High***

Answer Key

Q1. (B) $\frac{N(N-1)}{2}$ **Q2.** (D) Scoliosis (It's a postural deformity, not part of the triad). **Q3.** (C) Vajrasana **Q4.** (B) 1924 **Q5.** (B) Vitamin D & K (Vitamins A, D, E, K are fat-soluble). **Q6.** (A) Brouha **Q7.** (C) Comminuted fracture **Q8.** (C) Newton's Third Law (Law of Action & Reaction) **Q9.** (B) To provide an opportunity for mass participation of students within the institution. **Q10.** (A) Kyphosis **Q11.** (C) 55 dB **Q12.** (C) Vitamin C **Q13.** (D) Flamingo Balance Test **Q14.** (B) Stroke Volume **Q15.** (B) 45° **Q16.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A). **Q17.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A). **Q18.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A).

Q19. Bye: A privilege given to a team (usually in the first round) exempting them from playing, allowing direct entry to the second round. **Seeding:** Placing strong teams strategically in fixtures so they do not face each other in early rounds. **Q20.** 1. Horse riding. 2. Keeping a pillow between the knees while standing or sleeping. 3. Performing Padmasana. **Q21.** 1. Improves digestion and removes constipation. 2. Helps release trapped abdominal gas (flatulence). 3. Reduces belly fat. **Q22.** Inclusion means providing equal opportunities and creating a supportive environment where CWSN can participate in physical activities and sports alongside mainstream students in a regular educational setting. **Q23. Sprain:** A soft tissue injury involving the severe stretching or tearing of ligaments at a joint (bone to bone). **Strain:** A soft tissue injury involving the tearing of a muscle or tendon (muscle to bone). **Q24.** The Centre of Gravity is the imaginary point around which the entire weight of the body is balanced. A lower Centre of Gravity significantly increases an athlete's stability, making them harder to topple (e.g., wrestling stance).

Q25. 1. **Boundary:** Intramural is played within the school boundaries; Extramural is played outside between different schools. 2. **Objective:** Intramural focuses on mass participation and recreation; Extramural focuses on competitive excellence. 3. **Level of Play:** Intramural involves regular students; Extramural involves the best-selected school teams. **Q26. Flat Foot:** The complete absence of the natural medial arch in the sole, causing the entire foot to touch the ground. **Symptoms:** Pain in the feet during prolonged standing/walking. **Corrective measures:** 1. Walking on heels and toes. 2. Skipping rope or picking up marbles with toes. **Q27.** 1. **Extreme Caloric Restriction:** Starving reduces the Basal Metabolic Rate (BMR) and causes severe weakness. 2. **Skipping Meals:** Leads to a lack of sustained energy and overeating later. 3. **Dehydration:** Relying on liquid diets or restricting water drops performance and harms kidneys. **Q28. Procedure:** The subject steps up and down on a bench (20 inches for men, 16 for women) at a rate of 30 steps/minute for 5 minutes. The pulse is counted during recovery. **Short Formula:**
$$\frac{100 \times \text{Test duration in seconds}}{5.5 \times \text{Pulse count of 1-1.5 Min after Exercise}}$$
 Q29. Fast-Twitch (White) Fibers: Contract rapidly with explosive force but fatigue quickly; a higher percentage is essential for speed/strength events (e.g., 100m sprint). **Slow-Twitch (Red) Fibers:** Contract slowly but are highly resistant to fatigue; a higher percentage is essential for aerobic endurance (e.g., Marathon). **Q30. Class I (Fulcrum in middle):** Nodding the head at

the neck joint. **Class II (Load in middle):** Calf raise / standing on toes (Load is body weight).

Class III (Effort in middle): Bicep curl (Effort is bicep muscle).

Q31. (i) Knock-out tournament. (ii) $35 - 1 = 34$ matches. (iii) Next highest power of 2 is 64 (2^6), so 6 rounds. (iv) Byes = $64 - 35 = 29$ byes. **OR** Formula: $\frac{N_b+1}{2}$ (where N_b is the total number of byes).

Q32. (i) Rikli & Jones Senior Citizen Fitness Test. (ii) Chair Sit and Reach Test. (iii) Arm Curl Test. (iv) Agility and dynamic balance. **OR** 8 lbs (pounds). **Q33.** (i) Hearing and speech impairment. (ii) Inclusive Education / Inclusion. (iii) International Committee of Sports for the Deaf (ICSD) / Deaflympics. (iv) Use visual cues (flags or flashing lights) instead of whistles. **OR** 55 dB in the better ear.

Q34. League Tournament: A format where every team plays against every other team regardless of winning or losing. Matches = $\frac{N(N-1)}{2} = \frac{9 \times 8}{2} = 36$ matches. Rounds = $N = 9$ rounds. **Fixture (Cyclic):** Team 'Bye' is fixed at the top right. Other teams rotate clockwise around the fixed Bye across 9 rounds. (*Evaluator notes: Proper fixture drawn with 'Bye' static and teams 1-9 rotating clockwise*). **Q35.** The Triad is a syndrome of three interrelated conditions in female athletes driven by low energy availability. 1. **Eating Disorders:** Extreme dieting (Anorexia/Bulimia) to maintain low weight. 2. **Amenorrhea:** Absence of the menstrual cycle for 3+ months due to stress and low energy. 3. **Osteoporosis:** Weakening of bone density due to low estrogen and calcium. **Prevention:** Maintain a balanced diet matching caloric expenditure, ensure adequate calcium/vitamin D intake, and avoid overtraining. **Q36. Pre-competition:** Eat 3-4 hours before the event. Diet should be high in complex carbohydrates (to maximize glycogen), moderate in protein, and very low in fat/fiber to prevent stomach upset. Hydrate well. **During-competition:** Goal is maintaining hydration and blood sugar. Consume water, electrolyte drinks, and simple carbohydrates (glucose) in small, frequent amounts. **Post-competition:** Within 2 hours, consume high-quality proteins to repair muscle tissue micro-tears, and carbohydrates to rapidly replenish depleted glycogen stores, along with plenty of fluids for rehydration. **Q37.** Friction is the opposing force between two surfaces in contact. **Static Friction:** Resistance before an object starts moving (highest force). **Sliding Friction:** Resistance when an object slides over a surface (e.g., ice skating). **Rolling Friction:** Resistance when an object rolls (e.g., cycling). **Increasing Friction:** Using spikes on running shoes, applying chalk powder on gymnast's hands for better grip. **Decreasing Friction:** Applying wax to skis, polishing a carrom board, or wearing aerodynamic/smooth sportswear.